

DATS 6303: Final Project Proposal

Group 6

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Static American Sign Language (ASL) Alphabet Recognition using Deep Learning Techniques

1. What problem did you select, and why did you select it?

This project focuses on building a computer vision system that recognizes static American Sign Language (ASL) alphabet gestures from images. The goal is to improve accessibility by creating a model that can classify hand signs used in ASL alphabet communication and help bridge part of the communication gap between Deaf and Hard of Hearing users and hearing non-signers.

2. What database/dataset will you use? Is it large enough to train a deep network?

o Dataset: [Kaggle -ASL Alphabet](https://www.kaggle.com/grassknoted/aslalphabet_akashnagaraj_2018)

- i. @misc{https://www.kaggle.com/grassknoted/aslalphabet_akashnagaraj_2018,
- ii. title={ASL Alphabet},
- iii. url={<https://www.kaggle.com/dsv/29550>},
- iv. DOI={10.34740/KAGGLE/DSV/29550},
 - o Size: 87,000 images, which are 200x200 pixels across 29 classes (A-Z, plus Space, Delete, and Nothing).
 - o This data set definitely offers the robustness needed for a deep learning network

3. What deep network will you use? Will it be a standard form of the network, or will you have to customize it?

- Network: MobileNetV2 a CNN architecture
- We will have some customization, specifically, we will use the standard MobileNetV2 architecture pre-trained on ImageNet (Transfer Learning).
- Additionally, add a custom "head" comprising a 2D pooling layer, a Dropout layer (to prevent overfitting), and a final Dense layer with 29 neurons and a Softmax activation.

4. What framework will you use to implement the network? Why?

- Framework: Tensorflow/Keras
- Why? Tensorflow supports transfer learning with [tf.keras.applications](#). Additionally, we hope to use Streamlit to show a demo. With this framework, it also offers seamless integration

5. What reference materials will you use to obtain sufficient background on applying the chosen network to the specific problem that you selected?

- Reference Papers
 - i. Alsharif, Bader et al. "Deep Learning Technology to Recognize American Sign Language Alphabet." Sensors (Basel, Switzerland) vol. 23,18 7970. 19 Sep. 2023, doi:10.3390/s23187970
 - ii. Baihan, A., Alutaibi, A.I., Alshehri, M. et al. Sign language recognition using modified deep learning network and hybrid optimization: a hybrid optimizer (HO) based optimized CNNsSa-LSTM approach. Sci Rep 14, 26111 (2024).
<https://doi.org/10.1038/s41598-024-76174-7>
 - iii. G. Sravya, S. Das, K. Anjali, G. V. Priya and S. Gajula, "A Hybrid Deep Learning Approach for Sign Language Recognition Using MobileNetV2 and Attention Mechanism," 2026 Sixth International Conference on Advances in Electrical, Computing, Communications and Sustainable Technologies (ICAECT), Bhilai, India, 2026, pp. 1-6, doi: 10.1109/ICAECT68478.2026.11426032.
- Website
 - i. <https://keras.io/api/applications/mobilenet/>
 - ii. <https://pypi.org/project/streamlit/>

- Video Reference:
<https://www.youtube.com/watch?v=w8Qx40tHeEM&t=52s>

6. How will you judge the performance of the network? What metrics will you use?

- Primary Metric: Accuracy (overall correctness).
- Detailed Metrics: Precision, Recall, and F1-Score per class. This is crucial because some letters (like 'M', 'N', and 'S') look very similar and may have high error rates.
- Visualization: A Confusion Matrix to identify specifically which letters are being swapped by the model.

7. Rough Schedule

- Week 1 → Data & Preprocessing
- Week 2 → Model Training
- Week 3 → The Demo (Streamlit App Development)